

LECTURE / SESSION¹ PLAN

Name Lecturer: Sofia Gil-Clavel	Date session: 8	Expected number of students: 20
Course title	R-Workshop	
Topic of lecture or session	Text Mining	
Situational factors (e.g. group size, prior knowledge, expected motivation)	The class lasts 3hrs. The students are in different career stages, from PhD students to professors. The students have followed the previous sessions of the course or have experience with the topics. The students use different Operating Systems (Windows, Mac, or Linux).	
Intended learning outcomes of this session	At the end of the session the students will be able to: 1. Explain the main purpose of text mining. 2. Quantify what a document is about by calculating the tf-idf. 3. Extract relationships between words using the n-grams. 4. Calculate the n-grams tf-idf. 5. Calculate the correlation between words. 6. Visualize the relations between the words using networks.	
Learning material (book, chapters, ...)	The lecture is based on the book: • Robinson, Julia Silge and David. <i>Welcome to Text Mining with R Text Mining with R</i>. Accessed March 12, 2025. https://www.tidytextmining.com/.	
Media, equipment, tools	The students use their own laptops. The teacher needs access to a projector and a whiteboard.	

¹ A session means a teaching and learning session for university students in the bachelor or master. It can be a lecture, a seminar or a specific type of educational meeting with a group of students that is relevant for the discipline where the lecturer can demonstrate a whole range of teaching skills. It means that in one session each of the six didactic elements appears at least once. Your lesson plan should show that you are using a powerful learning environment and that students are activated.

Preparation for students	The students have access to the slides and codes before the class: https://github.com/SofiaG11/R_Course/tree/master/08_Session_TextMining
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Time ² (min.)	Didactic element (goal) ³ and topic ⁴	What the teacher does ⁵ (teacher activity)	What students do (learner activity)	Evaluation ⁶ (feedback/assessment)
20	Quick recap of previous session.	- The teacher uses the slides to remind the students what they learned the previous session: - How to open and clean textual data. - Transform textual data into a data frame. - Tidy text principles - Indexing - The teacher opens RStudio and opens the already written script that will be used during the class.	- Before telling each element the students learned, the teacher waits some seconds for the students to fill out the information out loud. - The students open and follow the steps that the teacher is explaining.	The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.
10	Explain the main purpose of text mining.	The teacher uses the slides to introduce the topic: Text Mining.	The students passively digest what the teacher is explaining.	The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these

² Indicate the planned duration in minutes

³ State the number(s) of the relevant ILO at each didactic element

⁴ Only key words

⁵ Specify the type of activity and write down also the questions that you prepared to ask in the session

⁶ Specify the type of evaluation, i.e. the way in which you assess if the objective(s) has/have been achieved

				confusions the teacher will be able to correct the student.
20	Quantify what a document is about by calculating the tf-idf.	The teacher uses the slides to explain what tf-idf and how to calculate it using R.	- The students passively digest what the teacher is explaining. - The students independently start filling out the missing parts of the R-script. This is based on what the teacher explained using the slides.	The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher uses the whiteboard to clarify.
20	Extract relationships between words using the n-grams.	The teacher uses the slides to explain what an n-gram is and why they are useful in text mining.	- The students passively digest what the teacher is explaining. - The students independently start filling out the missing parts of the R-script. This is based on what the teacher explained using the slides.	The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher uses the whiteboard to clarify.
20	Calculate the n-grams' tf-idf.	The teacher uses the slides to remind students what a tf-idf is and also explains how interpret it.	- The students passively digest what the teacher is explaining. - The students independently start filling out the missing parts of the R-script. This is based on what the teacher explained using the slides.	The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the

				teacher uses the whiteboard to clarify.
20	Calculate the correlation between words.	The teacher uses the slides to explain how to calculate the correlation between words and also explains how interpret it.	- The students passively digest what the teacher is explaining. - The students independently start filling out the missing parts of the R-script. This is based on what the teacher explained using the slides.	The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher uses the whiteboard to clarify.
20	Visualize the relations between the words using networks.	The teacher uses the slides to explain what a network is and why they are useful to visualize relations between words.	- The students passively digest what the teacher is explaining. - The students independently start filling out the missing parts of the R-script. This is based on what the teacher explained using the slides.	The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher uses the whiteboard to clarify.
20	Quick overview of Natural Language Processing	- The teacher uses the slides to explain what Natural Language Processing is and its differences with text mining. - The teacher provides some code so the students can play around with some basic Natural Language Processing functions.	- The students passively digest what the teacher is explaining. - The students independently start running the R-script. This is based on what the teacher explained using the slides.	The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the

				teacher uses the whiteboard to clarify.
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